LED (Addressable) Device-Items structure

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Resources

Features list - https://kno.wled.ge/features/effects/

Json format - wled_json_structure.md

Device

We will have 2 types of devices. Main device, that will control segments creation and SEGMENT_0 led behavior. And for each segment with id>0 will be created child device to control it's led behavior.

Device structure without additional segment (answer to hub.devices.list command)


```
{
    "api": "1.0",
    "error": null,
    "id": "<request_id>",
    "result": {
        "devices": [
            {
                "_id": "_DEVICE_ID_",
                "batteryPowered": false,
                "category": "led_strip",
                "deviceTypeId": "ezlopi",
                "gatewayId": "_GATEWAY_ID_",
                "name": "Led Strip Addressable",
                "parentDeviceId": "",
                "reachable": true,
                "armed": false,
                "serviceNotification": false,
                "roomId": "",
                "subcategory": "addressable_main",
                "type": "device",
                "security": "no",
                "status": "idle"
            }
        ]
    }
}
```

Device structure with additional segment (answer to hub.devices.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "<request_id>",
    "result": {
        "devices": [
            {
                "_id": "_DEVICE_ID_",
                "batteryPowered": false,
                "category": "led_strip",
                "deviceTypeId": "ezlopi",
                "gatewayId": "_GATEWAY_ID_",
                "name": "Led Strip Addressable",
                "parentDeviceId": "",
                "reachable": true,
                "armed": false,
                "serviceNotification": false,
                "roomId": "",
                "subcategory": "addressable_main",
                "type": "device",
                "security": "no",
                "status": "idle"
            },
            {
                "_id": "_DEVICE_ID_",
                "batteryPowered": false,
                "category": "led_strip",
                "deviceTypeId": "ezlopi",
                "gatewayId": "_GATEWAY_ID_",
                "name": "Led Strip Addressable Segment 1",
                "parentDeviceId": "_PARENT_DEVICE_ID_",
                "reachable": true,
                "armed": false,
                "serviceNotification": false,
                "roomId": "",
                "subcategory": "addressable_segment",
                "type": "device",
                "security": "no",
                "status": "idle",
                "info": {
                    "segment_id": "1"
                }
            },
            ...
        ]
    }
}
```

Device items:

Device name	Device category	Device subcategory	Item name	Item value type	Item element type	Mandatory
Led Strip Addressable	led_strip	addressable_main	led_segments	dictionary	led_segment	+
			led_effects	dictionary	led_effect	
			led_palettes	dictionary	string	
			switch	bool		
			dimmer	int		
			color_effect_select	int		
			color_palette_select	int		

			color_segment_grouping	int		
			color_segment_spacing	int		
			color_segment_offset	int		
			color_segment_revers	bool		
			color_segment_mirror	bool		
			color_effect_speed	int		
			color_effect_intencity	int		
			color_effect_custom_slider_1	int		
			color_effect_custom_slider_2	int		
			color_effect_custom_slider_3	int		
			color_effect_option_1	bool		
			color_effect_option_2	bool		
			color_effect_option_3	bool		
			color_segment_primary_color	rgb		
			color_segment_secondary_color	rgb		
			color_segment_ternary_color	rgb		
`Led Strip Addressable Segment 1`	`led_strip`	`addressable_segment`	switch	bool		-
			dimmer	int		
			color_effect_select	int		
			color_palette_select	int		
			color_segment_grouping	int		
			color_segment_spacing	int		
			color_segment_offset	int		
			color_segment_revers	bool		
			color_segment_mirror	bool		
			color_effect_speed	int		
			color_effect_intencity	int		
			color_effect_custom_slider_1	int		
			color_effect_custom_slider_2	int		
			color_effect_custom_slider_3	int		
			color_effect_option_1	bool		
			color_effect_option_2	bool		
			color_effect_option_3	bool		
			color_segment_primary_color	rgb		
			color_segment_secondary_color	rgb		
			color_segment_ternary_color	rgb		
`Led Strip Addressable Segment 2`	`led_strip`	`addressable_segment`	...	...		-

Device settings:

Device name	Device category	Device subcategory	Device settings
Led Strip Addressable	led_strip	addressable_main	ip_address

Items' structures

Unique parent device items

led_segments

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "deviceName": "_DEVICE_NAME_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "led_segments",
                "show": true,
                "valueType": "dictionary",
                "elementType": "led_segment",
                "elementsMaxNumber": 10
                "value": {
                    "0": {"name": "Main", "start": 0, "end": 125},
                    "1": {"name": "Custom segment", "start": 15, "end": 45},
                    ...},
                "valueFormatted": ""
            }
        ]
    }
}
```

Value type structure:

Field	Mandatory	Description
<dictionary_id>	+	Segment identifier
`name`	-	Segment name
`start`	+	Segment start led
`end`	+	Segment end led (not included into segment)

Note: This item will always contain "Main" segment with all supported leds in it and ID=0. This segment should be read only.

Add new segment

Add new segment

```
{
    "method": "hub.item.dictionary.value.add",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "element": {
            "type": "led_segment",
            "value": {"name": "Custom segment", "start": 15, "end": 45}
        }
    }
}
```

Broadcast on adding

```
{
    "id": "ui_broadcast",
    "msg_subclass": "hub.item.dictionary.updated",
    "result": {
        "_id": "_ITEM_ID_",
        "fwTimestampMs": 16300011122244,
        "deviceId": "_DEVICE_ID_",
        "deviceName": "<DEVICE_NAME>",
        "deviceCategory" : "<DEVICE_CATEGORY>",
        "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
        "roomName" : "<DEVICE_ROOM_NAME>",
        "userNotification": false,
        "serviceNotification":false,
        "notifications": [ "12314324", "978343" ],
        "name": "led_segments",
        "operation":"added",
        "element": {
            "1": {"name": "Custom segment", "start": 15, "end": 45}
            }
        }
    }
}
```

Modify existing segment

Modify existing segment

```
{
    "method": "hub.item.dictionary.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "key" : "1",
        "element": {
            "type": "led_segment",
            "value": {"name": "Custom segment", "start": 15, "end": 45}
        }
    }
}
```

Brodcast when modified

```
{
    "id": "ui_broadcast",
    "msg_subclass": "hub.item.dictionary.updated",
    "result": {
        "_id": "_ITEM_ID_",
        "fwTimestampMs": 16300011122244,
        "deviceId": "_DEVICE_ID_",
        "deviceName": "<DEVICE_NAME>",
        "deviceCategory" : "<DEVICE_CATEGORY>",
        "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
        "roomName" : "<DEVICE_ROOM_NAME>",
        "userNotification": false,
        "serviceNotification":false,
        "notifications": [ "12314324", "978343" ],
        "name": "led_segments",
        "operation":"updated",
        "element": {
            "1": {"name": "Custom segment", "start": 15, "end": 45}
            }
        }
    }
}
```

Delete existing segment

Delete existing segment

```
{
    "method": "hub.item.dictionary.value.remove",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "key" : "1"
    }
}
```

Brodcast when removed

```
{
    "id": "ui_broadcast",
    "msg_subclass": "hub.item.dictionary.updated",
    "result": {
        "_id": "_ITEM_ID_",
        "fwTimestampMs": 16300011122244,
        "deviceId": "_DEVICE_ID_",
        "deviceName": "<DEVICE_NAME>",
        "deviceCategory" : "<DEVICE_CATEGORY>",
        "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
        "roomName" : "<DEVICE_ROOM_NAME>",
        "userNotification": false,
        "serviceNotification":false,
        "notifications": [ "12314324", "978343" ],
        "name": "led_segments",
        "operation":"removed",
        "element": {
            "1": {"name": "Custom segment", "start": 15, "end": 45}
            }
        }
    }
}
```

led_effects

This item will show all the supported effects with an additional information to each effect.

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "deviceName": "_DEVICE_NAME_",
                "hasGetter": true,
                "hasSetter": false,
                "name": "led_effects",
                "show": true,
                "valueType": "dictionary",
                "elementType": "led_effect",
                "value": {"1": {"name": "Some name", "description": "Some description", "headpieceUrl":
"https://some_url"}, ...},
                "valueFormatted": ""
            }
        ]
    }
}
```

Value type structure:

Field	Mandatory	Description
<dictionary_id>	+	Effect identifier
`name`	+	Effect name
`description`	-	Effect description
`headpieceUrl`	-	Lint to the effect miniature

led_palettes

This item will show all the supported pallete ids with their names.

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "deviceName": "_DEVICE_NAME_",
                "hasGetter": true,
                "hasSetter": false,
                "name": "led_palettes",
                "show": true,
                "valueType": "dictionary",
                "elementType": "string",
                "value": {"1": "Random Cycle", "2": "Primary Color", ...},
                "valueFormatted": ""
            }
        ]
    }
}
```

Segment led behavior items

switch

Thsi item is needed to turn on/off whole LED strip.

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "deviceName": "_DEVICE_NAME_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "switch",
                "show": true,
                "valueType": "bool",
                "value": false,
                "valueFormatted": "false"
            }
        ]
    }
}
```

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : true
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "_ITEM_ID_",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "userNotification": false,
    "parentDeviceId": "<OPTIONAL_PARENT_DEVICE_ID>",
    "serviceNotification":false,
    "syncNotification":false,
    "fwTimestampMs": 16300011122244,
    "notifications": [ "12314324", "978343" ],
    "name": "switch",
    "value": true
  }
}
```

dimmer

This item is needed to control led strip's brightness.

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "deviceName": "_DEVICE_NAME_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "dimmer",
                "show": true,
                "valueType": "int",
                "value": 0,
                "minValue": 0,
                "maxValue": 100,
                "valueFormatted": "0"
            }
        ]
    }
}
```

Note: Possible values 0-100 in percents.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 5
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "_ITEM_ID_",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "userNotification": false,
    "parentDeviceId": "<OPTIONAL_PARENT_DEVICE_ID>",
    "serviceNotification":false,
    "syncNotification":false,
    "fwTimestampMs": 16300011122244,
    "notifications": [ "12314324", "978343" ],
    "name": "dimmer",
    "value": 5
  }
}
```

color_effect_select

This item controls segment's active effect. ("fx" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "deviceName": "_DEVICE_NAME_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_select",
                "show": true,
                "valueType": "int",
                "value": -1,
                "valueFormatted": "-1"
            }
        ]
    }
}
```

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 5
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "_ITEM_ID_",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "userNotification": false,
    "parentDeviceId": "<OPTIONAL_PARENT_DEVICE_ID>",
    "serviceNotification":false,
    "syncNotification":false,
    "fwTimestampMs": 16300011122244,
    "notifications": [ "12314324", "978343" ],
    "name": "led_effect_select",
    "value": 5
  }
}
```

Possible values:

Value	Description
-1	No effect running
0-66	Effect with specified id is running

color_palette_select

This item controls segment's active palette. ("pal" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_palette_select",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": 0,
                "maxValue": _COUNT_
            }
        ]
    }
}
```

Note: Max value should be calculated based on the supported pallets count.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_palette_select",
    "value": 1
  }
}
```

color_segment_grouping

This item controls segment's "grouping" parameter. ("grp" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_segment_grouping",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": 0,
                "maxValue": 255
            }
        ]
    }
}
```

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_segment_grouping",
    "value": 1
  }
}
```

color_segment_spacing

This item controls segment's "spacing" parameter. ("spc" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_segment_spacing",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": 0,
                "maxValue": 255
            }
        ]
    }
}
```

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_segment_spacing",
    "value": 1
  }
}
```

color_segment_offset

This item controls segment's "offset" parameter. ("of" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_segment_offset",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": _-LEN+1_,
                "maxValue": _LEN_
            }
        ]
    }
}
```

Note: Min max values should be calculated based on the segment length.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_segment_offset",
    "value": 1
  }
}
```

color_segment_revers

This item controls segment's "revers" parameter. ("rev" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_segment_revers",
                "show": true,
                "value": false,
                "valueFormatted": "false",
                "valueType": "bool",
            }
        ]
    }
}
```

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : true
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_segment_revers",
    "value": true
  }
}
```

color_segment_mirror

This item controls segment's "mirror" parameter. ("mi" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_segment_mirror",
                "show": true,
                "value": false,
                "valueFormatted": "false",
                "valueType": "bool",
            }
        ]
    }
}
```

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : true
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_segment_mirror",
    "value": true
  }
}
```

color_effect_speed

This item controls effect's "speed" parameter. ("sx" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_speed",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": 0,
                "maxValue": 255
            }
        ]
    }
}
```

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_effect_speed",
    "value": 1
  }
}
```

color_effect_intencity

This item controls effect's "intencity" parameter. ("ix" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_intencity",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": 0,
                "maxValue": 255
            }
        ]
    }
}
```

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_effect_intencity",
    "value": 1
  }
}
```

color_effect_custom_slider_1

This item controls effect's "custorm slider 1" parameter. ("c1" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_custom_slider_1",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": 0,
                "maxValue": 255
            }
        ]
    }
}
```

Note: If active effect does not supports custom slider 1 (based on effects metadata), "show" field's value should be changed to false.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_effect_custom_slider_1",
    "value": 1
  }
}
```

color_effect_custom_slider_2

This item controls effect's "custorm slider 2" parameter. ("c2" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_custom_slider_2",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": 0,
                "maxValue": 255
            }
        ]
    }
}
```

Note: If active effect does not supports custom slider 2 (based on effects metadata), "show" field's value should be changed to false.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_effect_custom_slider_2",
    "value": 1
  }
}
```

color_effect_custom_slider_3

This item controls effect's "custorm slider 3" parameter. ("c3" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_custom_slider_3",
                "show": true,
                "value": 0,
                "valueFormatted": "0",
                "valueType": "int",
                "minValue": 0,
                "maxValue": 31
            }
        ]
    }
}
```

Note: If active effect does not supports custom slider 3 (based on effects metadata), "show" field's value should be changed to false.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : 1
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_effect_custom_slider_3",
    "value": 1
  }
}
```

color_effect_option_1

This item controls effect's "option 1" parameter. ("o1" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_option_1",
                "show": true,
                "value": false,
                "valueFormatted": "false",
                "valueType": "bool",
            }
        ]
    }
}
```

Note: If active effect does not supports option 1 (based on effects metadata), "show" field's value should be changed to false.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : true
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_effect_option_1",
    "value": true
  }
}
```

color_effect_option_2

This item controls effect's "option 2" parameter. ("o2" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_option_2",
                "show": true,
                "value": false,
                "valueFormatted": "false",
                "valueType": "bool",
            }
        ]
    }
}
```

Note: If active effect does not supports option 2 (based on effects metadata), "show" field's value should be changed to false.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : true
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_effect_option_2",
    "value": true
  }
}
```

color_effect_option_3

This item controls effect's "option 3" parameter. ("o3" in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_effect_option_3",
                "show": true,
                "value": false,
                "valueFormatted": "false",
                "valueType": "bool",
            }
        ]
    }
}
```

Note: If active effect does not supports option 3 (based on effects metadata), "show" field's value should be changed to false.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : true
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_effect_option_3",
    "value": true
  }
}
```

color_segment_primary_color

This item controls segment's "primary color" parameter. (position 0 of "col" field in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_segment_primary_color",
                "show": true,
                "value": {
                    "wwhite":10,
                    "red":10,
                    "green":10,
                    "blue":10
                }
                "valueFormatted": "...",
                "valueType": "rgb"
            }
        ]
    }
}
```

Note: Only four color segments are supported: "red", "green", "blue", and "wwhite". All of them are mandatory.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : {
            "wwhite":0,
            "red":255,
            "green":255,
            "blue":255
        }
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_segment_primary_color",
    "value": {
        "wwhite":0,
        "red":255,
        "green":255,
        "blue":255
    }
  }
}
```

color_segment_secondary_color

This item controls segment's "secondary color" parameter. (position 1 of "col" field in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_segment_secondary_color",
                "show": true,
                "value": {
                    "wwhite":10,
                    "red":10.
                    "green":10,
                    "blue":10
                }
                "valueFormatted": "...",
                "valueType": "rgb"
            }
        ]
    }
}
```

Note: Only four color segments are supported: "red", "green", "blue", and "wwhite". All of them are mandatory.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : {
            "wwhite":0,
            "red":255,
            "green":255,
            "blue":255
        }
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_segment_secondary_color",
    "value": {
        "wwhite":0,
        "red":255,
        "green":255,
        "blue":255
    }
  }
}
```

color_segment_ternary_color

This item controls segment's "ternary color" parameter. (position 2 of "col" field in json API)

Item structure (answer to hub.items.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "result": {
        "items": [
            {
                "_id": "_ITEM_ID_",
                "deviceId": "_DEVICE_ID_",
                "hasGetter": true,
                "hasSetter": true,
                "name": "color_segment_ternary_color",
                "show": true,
                "value": {
                    "wwhite":10,
                    "red":10.
                    "green":10,
                    "blue":10
                }
                "valueFormatted": "...",
                "valueType": "rgb"
            }
        ]
    }
}
```

Note: Only four color segments are supported: "red", "green", "blue", and "wwhite". All of them are mandatory.

Set item value

```
{
    "method": "hub.item.value.set",
    "id": "<request_id>",
    "params": {
        "_id" : "_ITEM_ID_",
        "value" : {
            "wwhite":0,
            "red":255,
            "green":255,
            "blue":255
        }
    }
}
```

Item value updated

```
{
  "id": "ui_broadcast",
  "msg_subclass": "hub.item.updated",
  "result": {
    "_id": "588b7eb528b12d03be86f36f",
    "deviceId": "<DEVICE_ID>",
    "deviceName": "<DEVICE_NAME>",
    "deviceCategory" : "<DEVICE_CATEGORY>",
    "deviceSubcategory" : "<DEVICE_SUBCATEGORY>",
    "roomName" : "<DEVICE_ROOM_NAME>",
    "name": "color_segment_ternary_color",
    "value": {
        "wwhite":0,
        "red":255,
        "green":255,
        "blue":255
    }
  }
}
```

Device settings' structures

ip_address

This swtting shows strip's local IP iddress.

Setting structure (answer to hub.device.settings.list command)

```
{
    "api": "1.0",
    "error": null,
    "id": "_ID_",
    "method": "hub.device.settings.list",
    "result": {
        "settings": [
            {
                "_id": "_SETTING_ID_",
                "description": {
                    "text": "The IP address of this instance."
                    "lang_tag": "addressable_led_ip_address_description"
                },
                "deviceId": "_DEVICE_ID",
                "label": {
                    "text": "IP Address",
                    "lang_tag": "addressable_led_ip_address_label"
                },
                "status": "synced",
                "hasSetter": false,
                "value": "127:0:0:1",
                "valueType": "string"
            }
        ]
    },
    "sender": {
        "conn_id": "11ea88a5-70d1-4ba3-9e0b-9059f80a0dff",
        "type": "ui"
    }
}
```